```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>

struct node
{
    int regno;
    char name[30];
    struct node *next;
};

struct node *head = NULL;

void addStudent()
{
    struct node *newnode, *temp;
    newnode = (struct node*)malloc(sizeof(struct node));
    printf("Enter registration number:");
    scanf("%d", &newnode->regno);
    printf("Enter student name:");
    scanf("%s", newnode->name);
    newnode->next = NULL;

    if (head == NULL)
    {
        head = newnode;
    }
    else
    {
        temp = head;
        while (temp->next != 0)
        {
            temp = temp->next;
        }
        temp->next = newnode;
    }
    printf("Student is added successfully\n");
}

void removeStudent()
{
    struct node *temp, *prevnode;
    int regno;
```

```
    temp = head;
    prevnode = NULL;

    while (temp != NULL && temp->regno != regno)
    {
        prevnode = temp;
        temp = temp->next;
    }

    if (temp == NULL)
    {
        printf("Student recorded to be deleted not found\n");
    }
    else if (temp == head)
    {
        head = head->next;
        free(temp);
        printf("Student record deleted successfully\n");
    }
    else
    {
        prevnode->next = temp->next;
        free(temp);
        printf("Student record deleted successfully\n");
    }
}

void display()
{
    struct node *temp;
    if (head == NULL)
    {
        printf("No student registered\n");
        return;
    }

    temp = head;
    printf("Registration number \tName\n");
    while (temp != NULL)
    {
        printf("%d\t\t%s\n", temp->regno, temp->name);
        temp = temp->next;
    }
```

```
void sort()
{
    struct node *currnode, *nextnode;
    int temp;
    char tempname[30];

    currnode = head;
    while (currnode != NULL)
    {
        nextnode = currnode->next;
        while (nextnode != NULL)
        {
            if (currnode->regno > nextnode->regno)
            {
                temp = currnode->regno;
                currnode->regno = nextnode->regno;
                nextnode->regno = temp;

                strcpy(tempname, currnode->name);
                strcpy(currnode->name, nextnode->name);
                strcpy(nextnode->name, tempname);
            }
            nextnode = nextnode->next;
        }
        currnode = currnode->next;
    }
    printf("List sorted successfully\n");
}

void reverse()
{
    struct node *prevnode, *currnode, *nextnode;
    prevnode = 0;
    currnode = nextnode = head;
    while (nextnode != 0)
    {
        nextnode = nextnode->next;
        currnode->next = prevnode;
        prevnode = currnode;
        currnode = nextnode;
    }
    head = prevnode;
    printf("List reversed successfully\n");
```

```
int main()
{
    int choice;
    do
    {
        printf("\n-------------WORKSHOP REGISTRATION------------\n");
        printf("1. Add student\n");
        printf("2. Remove student\n");
        printf("3. Display student details\n");
        printf("4. Count total participants\n");
        printf("5. Sort list\n");
        printf("6. Reverse list\n");
        printf("7. Exit\n");
        printf("Enter your choice:");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                addStudent();
                break;
            case 2:
                removeStudent();
                break;
            case 3:
                display();
                break;
            case 4:
                count();
                break;
            case 5:
                sort();
                break;
            case 6:
                reverse();
                break;
            case 7:
                printf("Exiting program;\n");
                break;
            default:
                printf("Invalid choice\n");
        }
    } while (choice != 7);
```